

When Lines Become Contagious

Infectious Disease Modelling via Queueing Theory

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Motivation

When Lines
Become
Contagious

Newman

Probability you
get infected

Number of people
you infect

Fluctuations in
that number

Numerical results

$M/G/\infty$ Model

Extensions

References

- Current stochastic models assume homogeneity across populations.
- Research allowing for heterogeneity centers around simulations (Ying and O'Clery [9], Namilae et al. [6], Agnelli et al. [2], Johansson and Goscè [4], Ronchi and Lovreglio [8], Goscé et al. [3], Abadeer and Gorlatch [1], Rahn et al. [7].)
- In service systems, customers overlap rather than simultaneous interaction across the entire system.
- Current similar work (Kang et al. [5]) does not take the perspective of determining infection risk for individuals, and combining this perspective with a systemic view of individual infection numbers highlights underlying relationships in the model.
- Need for methodologies which allow for generalization of distributions.

Talk Outline

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- 5 $M/G/\infty$ Model
- 6 Extensions

Model

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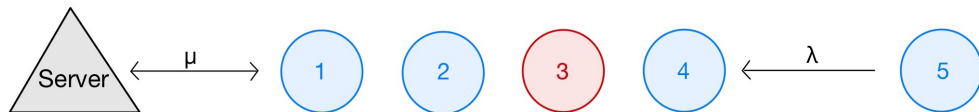
Fluctuations in
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Extensions

References



- Arrival rate λ , Poisson process
- Service rate μ , exponential
- FCFS discipline

We suppose an infectious probability p that any customer is already **infectious** upon entering the system, and a susceptible probability $1 - p$ that a customer is **healthy**.

Model

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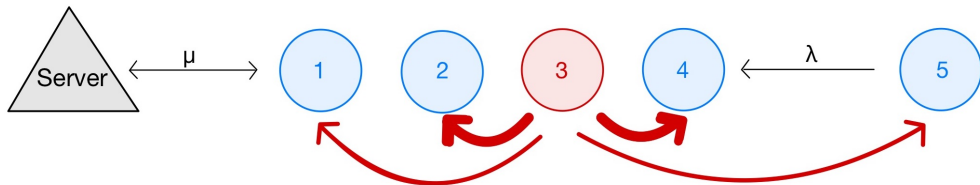
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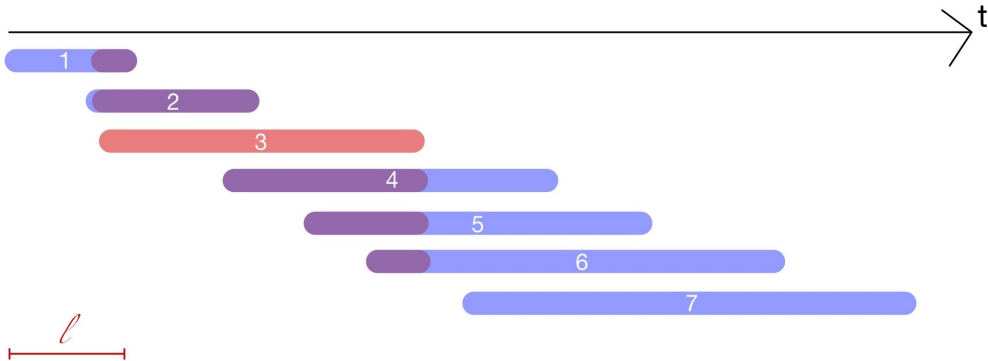
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We suppose that if a **susceptible** customer is in the system for at least ℓ units of time overlapping with an **infectious** customer, then the susceptible customer becomes infected.

Methodology

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For an arrival in steady state, we can partition all the customers into those who arrive before this arrival and those who arrive after.

- **Case 1 “Upon Arrival”** the customer interacts with the customers already in the queue upon their arrival.
- **Case 2 “After Arrival”** the customer interacts with the customers who arrive while they’re in the system.

These cases are related only through the shared amount of time the customer spends in the system.

Susceptible model

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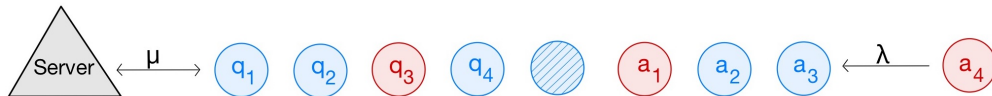
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- Probability p infectious
- Probability $1 - p$ susceptible
- Overlap threshold ℓ
- Center perspective for susceptible arrival
- Partition into upon and after arrivals

Susceptible model

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Notation

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Queue upon arrival:

- Q : queue length.
- S_i : service time for i^{th} customer in queue.
- R_i : remaining sojourn time for i^{th} customer in queue.

Susceptible arrival:

- S : their service time
- $\mathcal{S} = S + \sum_{i=1}^Q R_i$: their sojourn time.

Important events for susceptible arrival:

- I_{ua} : event they get infected by a customer already in the queue upon arrival.
- I_{aa} : event they get infected by an arrival after them.
- $I = I_{ua} \cup I_{aa}$: event they get infected while in the system.

Closest people

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Our case methodology makes it useful to hone in on specific overlaps:

After Arrivals

- consider only *infectious arrivals* after us: Let M_I count how many there are.
- for known sojourn time and M_I , arrival times distributed uniformly.
- the closest infectious person: the first infectious arrival, $A_{(1)}$.

Lemma 1

For k arrivals, sojourn time x , $\mathbb{P}(I_{aa}^c | M_I = k, \mathcal{S} = x) = \mathbb{P}(A_{(1)} > x - \ell | M_I = k) = \left(\frac{\ell}{x}\right)^k$.

Upon Arrivals

- consider all earlier arrivals as customers $1, \dots, k$
- the closest infectious person: the last infectious arrival, $I_{\max} \leq k$

Lemma 2

Supposing queue length k , for all $1 \leq j \leq k$, $\mathbb{P}(I_{\max} = j | Q = k) = p(1 - p)^{k-j}$.

Upon arrivals: probability they can infect us

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Lemma 3

Given queue length k and for some $1 \leq j \leq k$ the j^{th} is the last infectious customer in the queue, the probability that a susceptible arrival gets infected is equal to

$$\mathbb{P}(I_{ua} | I_{\max} = j, Q = k) = \sum_{n=0}^{j-1} e^{-\mu\ell} \frac{(\mu\ell)^n}{n!}.$$

Sequential dependency of FCFS means requiring $I_{\max}$ to have remaining sojourn time at least ℓ also ensures they overlap with us at least ℓ , so

$$\mathbb{P}(I_{ua} | I_{\max} = j, Q = k) = \mathbb{P}\left(\sum_{m=1}^j R_m \geq \ell\right)$$

Upon arrivals: symmetrical probabilities

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Theorem 4

Let a single susceptible customer join the $M/M/1$ queue in steady state, then the probability the customer gets infected by a customer who is in queue upon their arrival is equal to the probability they get infected by a customer who joins the queue after their arrival,

$$\mathbb{P}(I_{ua}) = \mathbb{P}(I_{aa}) = \frac{\lambda p}{\mu - \lambda + \lambda p} e^{-(\mu - \lambda)\ell}. \quad (1)$$

- For a random threshold, one can then integrate over the distribution.
- Note that first term is $\mathbb{P}(S_{\mu - \lambda} > S_{\lambda p}) = \frac{\lambda p}{\mu - \lambda + \lambda p}$.
- Note that second term is $\mathbb{P}(S > \ell) = e^{-(\mu - \lambda)\ell}$.

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Let $I = I_{ua} \cup I_{aa}$ be the event that a susceptible customer gets infected while in the system.

Theorem 5

Suppose a susceptible customer joins the $M/M/1$ queue in steady state, then the probability the customer gets infected is equal to

$$\mathbb{P}(I) = \mathbb{P}(I_{ua} \cup I_{aa}) = \frac{2\lambda p}{\mu - \lambda + 2\lambda p} e^{-(\mu - \lambda)\ell}.$$

Again, intuitive product of two probabilities:

- the exponential of pass-through rate is more than the exponential of twice the infectious arrival rate: twice the rate since equal probability of infection both upon and after arrival,
- and sojourn time is at least infection threshold.

Experiment: exactly k transmissions

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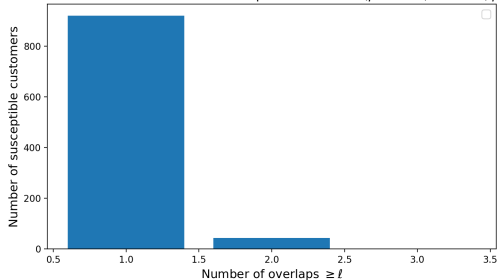
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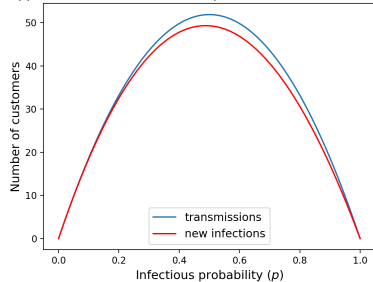
Extensions

References

Number of transmissions each infected person received ($\rho = 0.05, \ell = 0.25, p = 0.4$)



Approximate number infected ($\rho = 0.05, \ell = 0.25, t = 50000$)



963 susceptible people became infected. Experimentally, they were infected by exactly one person 95.43% of the time.

Experiment: exactly k transmissions

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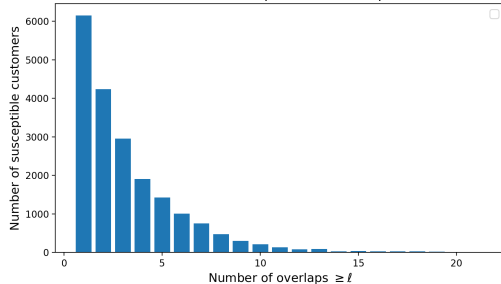
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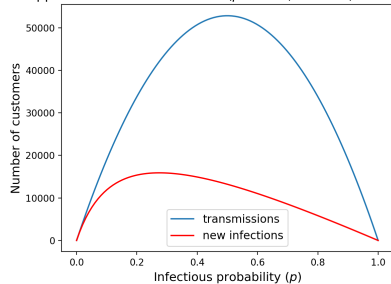
Extensions

References

Number of transmissions each infected person received ($\rho = 0.75, \ell = 0.25, p = 0.4$)



Approximate number infected ($\rho = 0.75, \ell = 0.25, t = 50000$)



19,904 susceptible people became infected. Experimentally, they were infected by exactly one person 30.88% of the time.

Reframe problem: dual perspective

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- Center perspective for **infectious arrival**
- Partition into upon and after arrivals

Upon arrivals: closest overlap vs. counting overlaps

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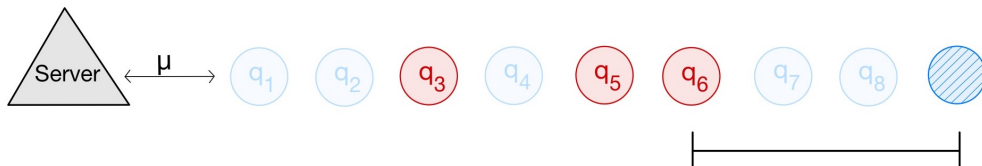
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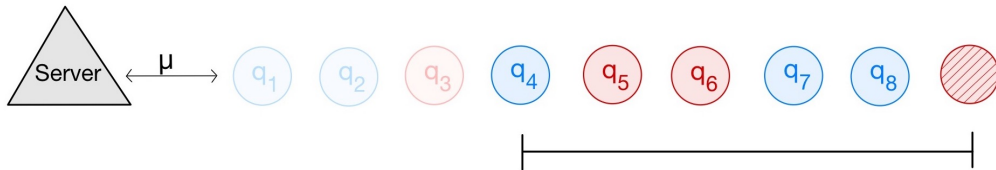
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Susceptible arrival:



Infectious arrival:



Reconsider overlaps

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Our case methodology again makes it useful to hone in on specific overlaps:

After Arrivals

- consider only the susceptible arrivals after us
- the furthest susceptible customer: last arrival while we've still got ℓ left in the system

Upon Arrivals

- consider all earlier arrivals as customers $1, \dots, k$.
- some of these customers may have remaining sojourn time $< \ell$.
- the furthest customer potentially infectible: first customer with remaining sojourn time at least ℓ .
- $\mathbf{1}_{\tilde{s}_j > \ell}$: let this be the event customer j is *first* in queue with sojourn time $\geq \ell$.

Lemma 6

Supposing queue length k , for all $0 < j \leq k$, $\mathbb{P}\left(\mathbf{1}_{\tilde{s}_j > \ell} | Q = k\right) = \frac{(\mu\ell)^{j-1} e^{-\mu\ell}}{(j-1)!}$.

Upon arrivals: counting number you infect

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Let N_{ua} be the number of susceptible customers in the queue upon arrival you infect (so $N_I = N_{ua} + N_{aa}$ is the total number of new infections you cause).

Lemma 7

Supposing queue length r and for some $j \leq r$ that the j^{th} customer is the first in the line with sojourn time at least ℓ , then for any $k \geq 1$ where $k \leq r - j + 1$, the probability infect k of these customers is

$$\mathbb{P}\left(N_{ua} = k | Q = r, \mathbf{1}_{\tilde{S}_j > \ell}\right) = \binom{r-j+1}{k} (1-p)^k p^{r-j+1-k}.$$

- Only persons $j, \dots, r$ can get infected.
- Exactly k of them must be susceptible.
- Binomial with parameters $(r - j + 1)$ and $(1 - p)$.

Number of people you infect

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Theorem 8

For an infectious arrival to the $M/M/1$ queue in steady state, and for any $k \geq 1$,

$$\mathbb{P}(N_{ua} = k) = \mathbb{P}(N_{aa} = k) = \frac{\mu - \lambda}{\mu - \lambda + \lambda(1 - p)} \left(\frac{\lambda(1 - p)}{\mu - \lambda + \lambda(1 - p)} \right)^k e^{-(\mu - \lambda)\ell}.$$

Further,

$$\mathbb{P}(N_I = k) = \frac{\mu - \lambda}{\mu - \lambda + 2\lambda(1 - p)} \left(\frac{2\lambda(1 - p)}{\mu - \lambda + 2\lambda(1 - p)} \right)^k e^{-(\mu - \lambda)\ell}.$$

- Swapping susceptible arrival rate $\lambda(1 - p)$ for infectious arrival rate of λp converts between number you infect to number who infect you.
- First term: no more than k infections.
- For $k = 1$, with swap, dropping exactly k term, recover results of Theorems 4 and 5.

Expectation

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Theorem 9

For a single infectious arrival to the $M/M/1$ queue in steady state, the expected number of susceptible customers who they infect is

$$\mathbb{E}[N_I] = \frac{2\lambda(1-p)}{\mu-\lambda} e^{-(\mu-\lambda)\ell}.$$

- Invariant to scaling $\rho = \frac{\lambda n}{\mu n}$, as long as we scale ℓ by $1/n$.
- Matches results by Kang et al. [5] when ℓ exponential.

Number of new infections in time t

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- $t\lambda(1-p)$ susceptible arrivals and $t\lambda p$ infectious arrivals
- Expect $t\lambda(1-p)\mathbb{P}(I)$ new infections
- Expect infectious arrivals to transmit infection $t\lambda p\mathbb{E}[N_I]$ times

Counts multiple transmissions to the same susceptible customer!

- $\frac{\lambda}{\mu} \ll 1$: number of overlaps $\downarrow$, repeat transmissions goes to 0.
- $\frac{\lambda}{\mu} \rightarrow 1$: duplicate transmissions $\uparrow$.

So we get ratio of $\frac{\text{New Infections}}{\text{Total Number of Transmissions}} = \frac{\mu - \lambda}{\mu - \lambda + 2\lambda p} = 1 - \frac{2\lambda p}{\mu - \lambda + 2\lambda p}$.

- As $p \rightarrow 0$, ratio goes to 1
- As $p \rightarrow 1$, ratio goes to $\frac{\mu - \lambda}{\mu + \lambda}$

Number of new infections

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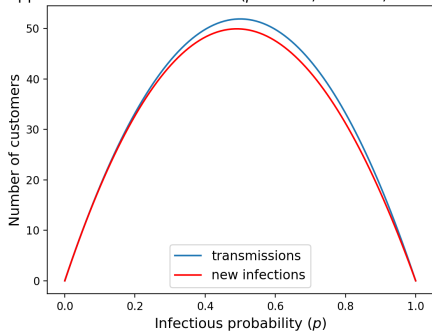
Numerical results

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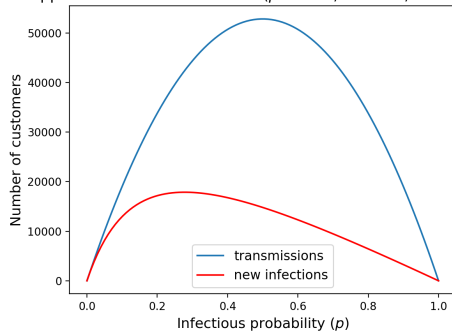
Extensions

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Approximate number infected ($\rho = 0.05$, $\ell = 0.25$, $t = 50000$)



Approximate number infected ($\rho = 0.75$, $\ell = 0.25$, $t = 50000$)



Higher moments: geometric component

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Theorem 10

Let N_I be the number of susceptible customers that are infected by one infectious customer, then the m^{th} moment of N_I is given by

$$\mathbb{E}[N_I^m] = \frac{\mu - \lambda}{\mu - \lambda + 2\lambda(1 - p)} \text{Li}_{-m} \left(\frac{2\lambda(1 - p)}{\mu - \lambda + 2\lambda(1 - p)} \right)$$

where $\text{Li}_m(x)$ is the polylogarithm function and is defined by

$$\text{Li}_m(x) = \sum_{j=1}^{\infty} \frac{x^j}{j^m}.$$

Variance

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Theorem 11

Consider a single infectious arrival to the $M/M/1$ queue, then the variance of the number of new infections arising while this customer is in the system is,

$$\text{Var}[N_I] = \frac{2\lambda(1-p)(\mu - \lambda + 4\lambda(1-p))}{(\mu - \lambda)^2} e^{-(\mu - \lambda)\ell} - \frac{4\lambda^2(1-p)^2}{(\mu - \lambda)^2} e^{-2(\mu - \lambda)\ell}.$$

Numerical results: susceptible arrival

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We fix $\mu = 1$ and $\ell = 0.5$.

- Average over 5 simulations
- 100,000 customers per simulation
- Note as $\frac{\lambda}{\mu} \rightarrow 1$, the number of customers to reach steady state increases

λ	p	probability infected (formula)	probability infected (simulation)
.50	.01	.01527	.01578
.75	.01	.04995	.04941
.90	.01	.14510	.14885
.50	.02	.02995	.03023
.75	.02	.09455	.09473
.90	.02	.25180	.24421
.50	.10	.12980	.12893
.75	.10	.33094	.33323
.90	.10	.61150	.60601
.50	.50	.38940	.38959
.75	.50	.66187	.66032
.90	.50	.85611	.85904

Numerical results: infectious arrival

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We fix $\mu = 1$ and $\ell = 0.5$.

- Average over 5 simulations
- 500,000 customers per simulation
- As $\frac{\lambda}{\mu} \rightarrow 1$, customers required for steady state increases

λ	p	$\mathbb{E}[N_I]$ (formula)	$\mathbb{E}[N_I]$ (simulation)	$\text{Var}[N_I]$ (formula)	$\text{Var}[N_I]$ (simulation)
.50	.01	1.54203	1.57283	5.27060	5.42214
.75	.01	5.24203	5.27382	40.03847	39.88854
.90	.01	16.95091	17.35411	333.74799	371.66641
.50	.02	1.52645	1.53207	5.18008	5.25931
.75	.02	5.18908	5.22471	39.28611	40.44939
.90	.02	16.77969	16.77969	327.20915	317.46625
.50	.10	1.40184	1.39161	4.48331	4.44239
.75	.10	4.76548	4.72859	33.52287	32.88926
.90	.10	15.40992	15.50204	277.22569	281.78650
.50	.50	0.77880	0.77984	1.72987	1.72464
.75	.50	2.64749	2.66862	11.52323	11.60138
.90	.50	8.56106	8.53275	89.36840	88.75154

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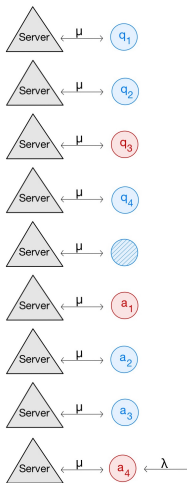
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- Probability p infectious
- Probability $1 - p$ susceptible
- Overlap threshold ℓ
- Partition into upon and after arrivals
 - Upon arrivals can leave *after us*
 - After arrivals can leave *before us*

$M/G/\infty$ Model

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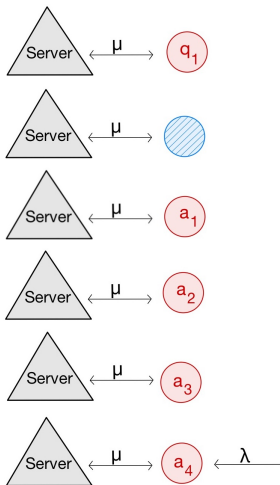
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- Probability $1 - p$ susceptible
- Overlap threshold ℓ
- Partition into upon and after arrivals
 - Upon arrivals can leave *after us*
 - After arrivals can leave *before us*

Worth noting:

- Lose the sequential relationship between arrivals and departures
- Upon and after arrivals now *both* reduce to only considering infectious customers

Loss of symmetry

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Theorem 12

Let a susceptible customer join the $M/G/\infty$ queue, then the probability they are infected upon arrival is

$$\mathbb{P}(I_{ua}(t)) = \overline{G}(\ell) \cdot \left(1 - e^{-q_I(t)\overline{G}_e(\ell)}\right).$$

The probability they are infected after arrival is

$$\mathbb{P}(I_{aa}(t)) = \overline{G}(\ell) - e^{\lambda p \overline{G}(\ell) \ell} \cdot \mathbb{E} \left[e^{-\lambda p \overline{G}(\ell) S} \cdot \mathbf{1}_{S \geq \ell} \right].$$

- $\mathbb{P}(I_{ua}(t))$ is the product of two probabilities:
 - probability sojourn time is at least ℓ .
 - thinned version of infectious customer: at least one has remaining service time at least ℓ .
- When service is exponential, $\mathbb{P}(I_{aa}(t)) = e^{-\mu \ell} \cdot \frac{\lambda p e^{-\mu \ell}}{\lambda p e^{-\mu \ell} + \mu}$.
 - probability sojourn time is at least ℓ .
 - sojourn time at least ℓ is more than arrival of first infectious customer with service at least ℓ .

Probability you get infected

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Theorem 13

Let $I(t) = I_{ua} \cup I_{aa}$ be the event that a susceptible customer gets infected while in the queue. Then, we have the following expression for probability of $I(t)$:

$$\mathbb{P}(I(t)) = \overline{G}(\ell) - e^{\lambda p \overline{G}(\ell) \ell} \cdot \mathbb{E} \left[e^{-\lambda p \overline{G}(\ell) S} \cdot \mathbf{1}_{S \geq \ell} \right] \cdot e^{-q_I(t) \overline{G}_e(\ell)}.$$

Moreover, when the service distribution is exponential, we have that

$$\mathbb{P}(I(t)) = e^{-\mu \ell} \left(1 - \frac{\mu}{\lambda p e^{-\mu \ell} + \mu} \cdot e^{-\left(\frac{\lambda p e^{-\mu \ell}}{\mu} (1 - e^{-\mu t}) \right)} \right).$$

Moments of number you infect

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Theorem 14

The moments of the number an infectious arrival to the $M/G/\infty$ queue infects are

$$\mathbb{E}[N_I^m(t)] = \mathbb{E} \left[T_m \left(q_S(t) \overline{G}_e(\ell) + \lambda(1-p)(S-\ell) \overline{G}(\ell) \right) \mathbf{1}_{S>\ell} \right],$$

where $T_m(x)$ is the m^{th} Touchard polynomial at x . These polynomials have the following polynomial expression

$$T_m(x) = e^{-x} \sum_{j=0}^{\infty} \frac{x^j j^m}{j!} = \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} x^j.$$

where $S(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ is the Stirling number of the second kind, representing the number of partitions of a set of size n into k disjoint non-empty subsets.

Comorbidity

When Lines
Become
Contagious

Newman

Probability you
get infected

Number of people
you infect

Fluctuations in
that number

Numerical results

$M/G/\infty$ Model

Extensions

References



$M/G/\infty$ Comorbidity:

- allow more classes of customers
- represents additional diseases in the system
- likelihood of contracting specific subset of diseases

Extensions

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References

- $M/M/1$ Cybersecurity
 - side-channel attacks
 - likelihood “malicious agent” after arrival overlaps with sufficient percentage of your *service*
 - worth finding: number of agents who steal info from you
 - For $M/G/\infty$: allow agents to “share info”

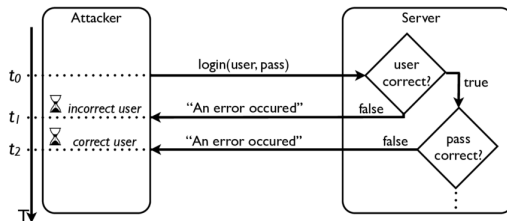


Figure: Unintentional and Hidden Information Leaks in Networked Software Applications - Scientific Figure on ResearchGate.

Conclusion

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References

- Upon and after arrival methodology provides blueprint for analysis
- Methodology allows us to generalize the distribution of the threshold
- Overlaps an exciting new way of exploring customer-to-customer behaviour in a queue.

Where next?

- Infected customers become infectious while still in the queueing system.
- Subclasses of both infectious and susceptible customers (e.g. masked or unmasked) with different infection thresholds between the classes.
- Spatial infection (e.g. measles) where a “spot” in line becomes infectious for a period of time.
- Allowing a “total contact” threshold measuring a customer’s continuous overlap with *any* infectious customers.
- Allowing a weighted threshold which is reduced if a customer simultaneously overlaps with multiple infectious customers.

References I

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Extensions

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When Lines
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